

CAREER EPISODE 2

Research on application of MFCCs in speech recognition

A. INTRODUCTION

CE 2.1

In my engineering course curriculum I executed the project “Research on application of MFCCs in speech recognition” and I submitted it as an academic project assignment. For the proficient conduct of current research, I investigated the speech recognition system thoroughly and gathered an understanding of the application of the MFCCs, however using the MATLAB platform I performed the simulink modeling analysis that I presented in an effective manner.

B. BACKGROUND

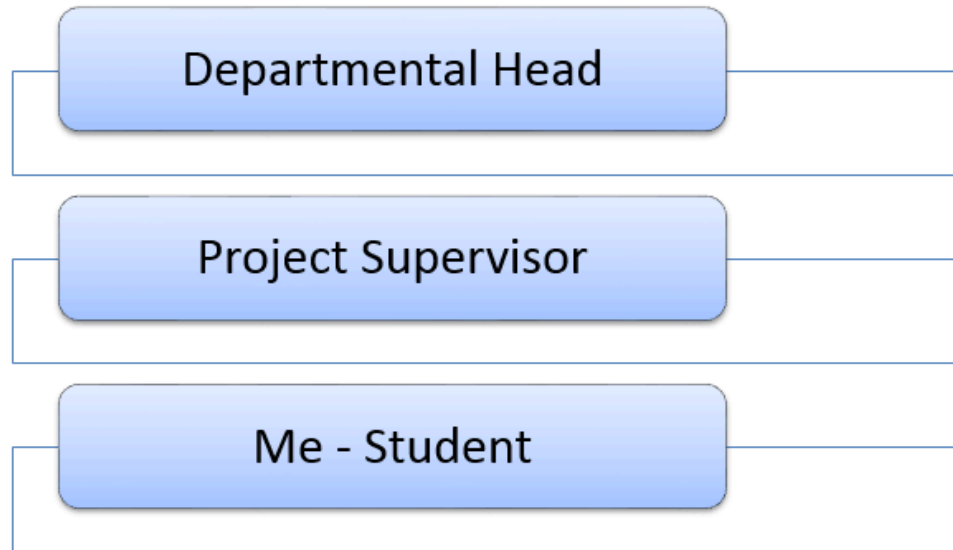
CE 2.2

By referring to the latest research work I conducted a wide review on the speech recognition process and the MFCCs application, however my focus was on the steps involved in the use of the MFCCs in the concerned system. So in the research work I referred to the research articles published recently in the indexed journals and moreover I briefly investigated the speaker recognition variants for the basic understanding whereas all the associated terms and systems were studied by me. I enlisted all the primary points concerning the automated speech recognition system and recorded them in sequence with proper details for reference. I worked hard using my research aptitude and reliable research sources which helped in accumulating desired information on my project topic. With all significant considerations I worked in MATLAB to develop a speech recognition model with MFCC application and analyzed the performance under certain conditions.

CE 2.3

- ◆ In my current research work I investigated the speech recognition system thoroughly and gathered an understanding of the application of the MFCCs, however using the MATLAB platform I performed the modeling analysis that I presented in an effective manner.
- ◆ Based on wide research I understood the significant steps involved in MFCC and further I optimally studied the application of MFCCs in the speech recognition system and acquired the desired information.
- ◆ I studied that MFCC is a feature which is widely deployed in automated speech as well as speaker recognition.
- ◆ Based on a research review I considered MFCC for the feature extraction in terms of characteristics constraints of words whereas under the MATLAB platform I modeled a speech recognition system and performance analysis conducted under certain conditions, whereas different blocks were configured as per the standard procedure.
- ◆ In order to analyze the performance of the modeled system the speech signal taken from an audio device and MATLAB read this input, however as per the modeling signal processed and respective characteristics constraints analyzed.
- ◆ In order to analyze the signal I utilized the spectrum analyzer at different blocks such as Audio device spectrum analyzer used for input audio device spectrum, FFT spectrum analyzer for FFT signal analysis, MFCC spectrum analyzer for MFCC analysis, DCT spectrum analyzer for DCT analysis, Audio weighting spectrum analyzer for its analysis, however proper configuration carried out to achieve the final output.

CE 2.4



C. PERSONAL ENGINEERING ACTIVITY

CE 2.5

Under my research part I investigated the processing steps involved in the existing speech recognition system and particularly in this research my focus was on the available algorithms for the recognition of speech signals. For example, based on the research I found that the initial step in such systems involves extraction of features which includes identification of the audio signal components which are desired to identify the linguistic content and further discard remaining stuff like the information-carrying emotions, background noise, and others. I investigated that the primary point for speech understanding is sound produced by humans gets filtered by vocal tract shape including teeth, tongue, and others. This shape helps determine the sound coming out and with the appropriate determination of shape, the phoneme being generated could be represented accurately. With respect to my engineering domain knowledge I also understood that the primary function of the MFCC is to represent the short-time power spectrum envelope. Therefore I further explored the concepts of MFCC as a feature i.e. widely deployed in automated speech as well as speaker recognition system, however in the research I found that before MFCC, linear prediction Cepstral coefficients were utilized for the ASR. Based on a research review I considered MFCC for the feature extraction in terms of characteristics constraints of audio signal whereas under the MATLAB platform I modeled a speech recognition system using distinct blocks of simulink toolbox.

CE 2.6

After basic understanding, I proceeded to understand about the steps involved in MFCC and its primary aspects. As per the standard procedure I reviewed the steps involved which I summarized here as-

1. Before the implementation of MFCC, framing of a speech signal was executed to represent the complete speech signal into several sub-frames.
2. Thereafter window function applied to each frame.
3. Thereafter FFT analysis to obtain the spectrum representation or frequency domain analysis.
4. Apply filter bank and compute the log of filter bank energies to get MFCC
5. Apply DCT for expressing a finite number of sequences in terms of summation of cos function at various frequency values.

By following above steps I modeled a speech recognition system to illustrate the MFCC application under MATLAB and the developed model I depicted here in figure 1-

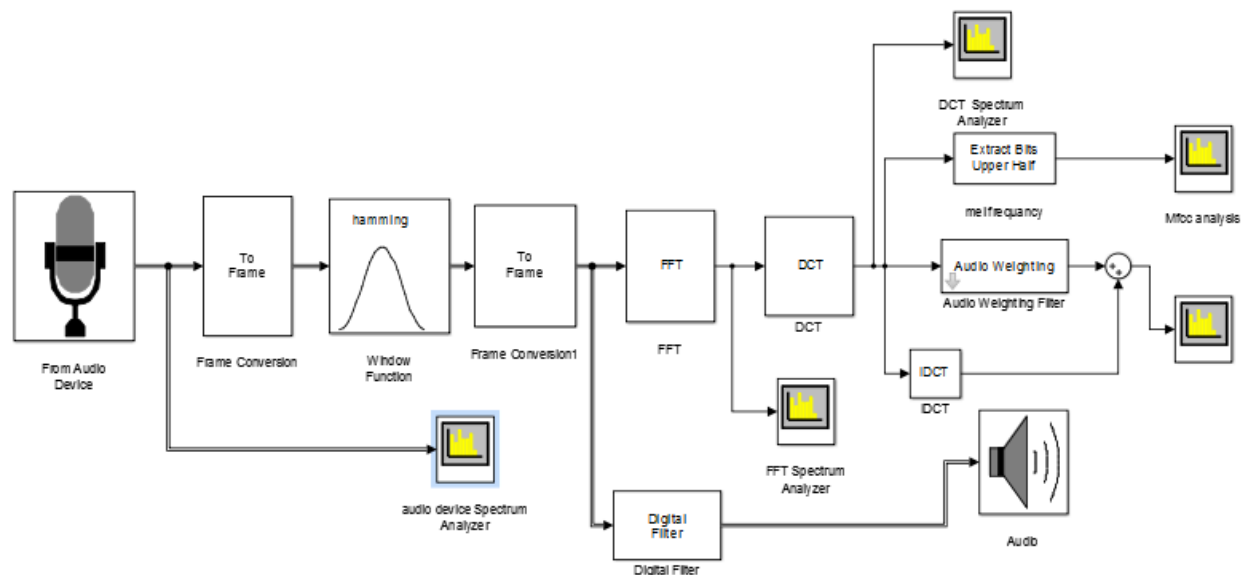


Fig. 1: Simulink model

As per the model I carried out the framing of the speech signals into the short frames, computation of the power spectrum's periodogram estimate for the respective frames, Mel filter bank application to power spectra and energy summing up in respective filters, and many other steps. While working on the simulink modeling I configured the selected blocks in terms of their technical characteristics within the Simulink toolbox of MATLAB. For example I first

configured the audio device block wherein figure 2 refers to the audio device that can send sound data to the audio device of PC and figure 3 refers to the record of audio signal from the audio device of PC-

The image shows the configuration window for the 'To Audio Device' block in Simulink. The title bar reads 'To Audio Device'. Below the title bar, there is a description: 'Send sound data to your computer's audio device.' The 'Parameters' section contains the following settings: 'Device' is set to 'No device available.' (indicated by a blue highlight and a dropdown arrow); 'Inherit sample rate from input' is unchecked; 'Sample rate (Hz)' is set to '44100'; 'Device data type' is set to 'Determine from input data type' (indicated by a dropdown arrow); 'Automatically determine buffer size' is checked; and 'Queue duration (seconds)' is set to '1.0'.

Fig. 2: Simulink block configuration

The image shows the configuration window for the 'From Audio Device' block in Simulink. The title bar reads 'From Audio Device'. Below the title bar, there is a description: 'Record sound data from your computer's audio device.' The 'Parameters' section contains the following settings: 'Device' is set to 'Primary Sound Capture Driver' (indicated by a dropdown arrow); 'Number of channels' is set to '2'; 'Sample rate (Hz)' is set to '44100'; 'Device data type' is set to 'Determine from output data type' (indicated by a dropdown arrow); 'Automatically determine buffer size' is checked; and 'Queue duration (seconds)' is set to '1.0'. The 'Outputs' section contains the following settings: 'Frame size (samples)' is set to '1024' and 'Output data type' is set to 'single' (indicated by a dropdown arrow).

Fig. 3: Simulink block configuration

In order to analyze the audio signal I also connected a block for audio device spectrum analyzer and therefore I configured different constraints as depicted in fig 4 & 5 to obtain the spectrum of the audio signal-

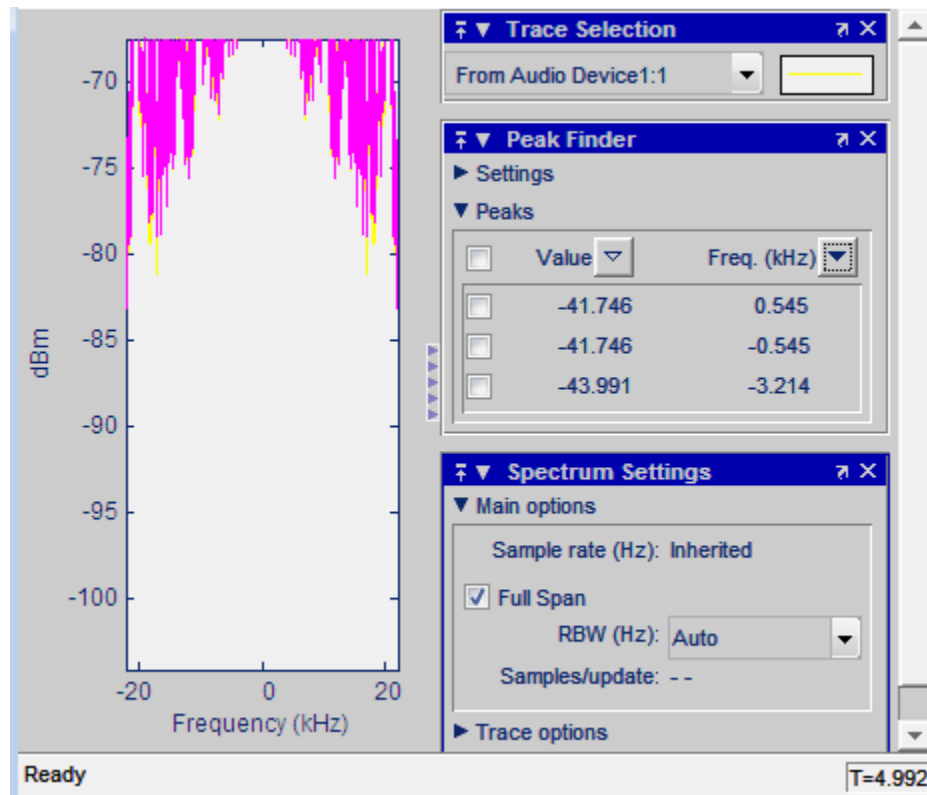


Fig. 4: Configuration part during Simulink modelling

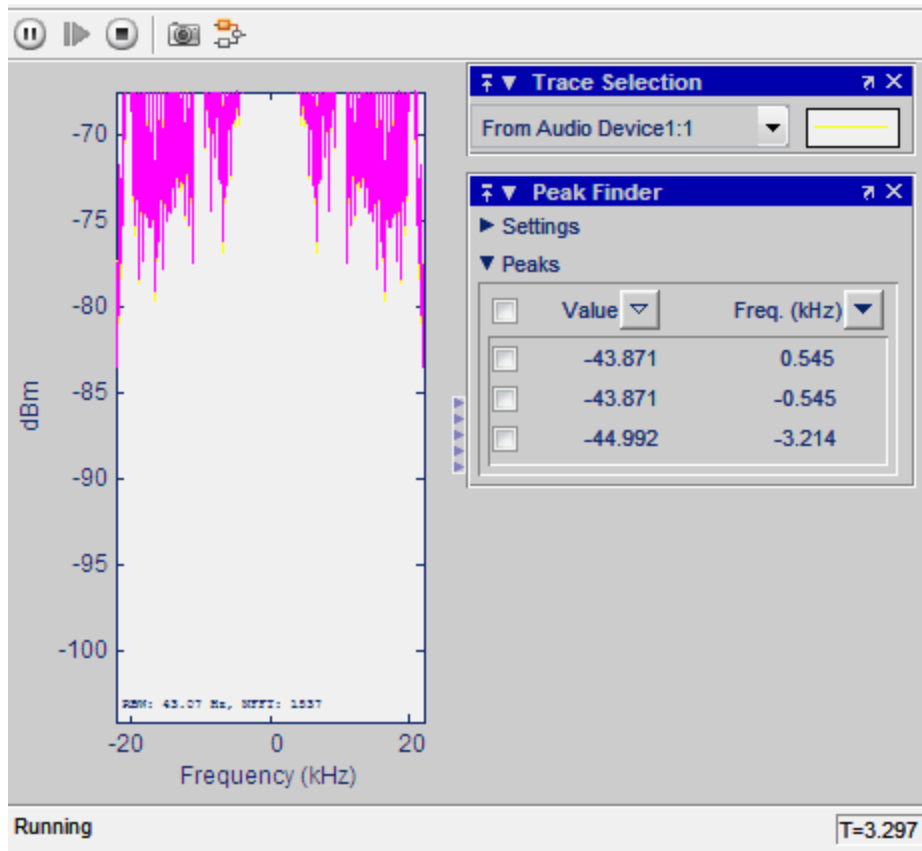


Fig. 5: Simulink block configuration

Thereafter I worked on frame conversion block to set the sampling mode as frame-based for the output signal as illustrated in Fig. 6 below-

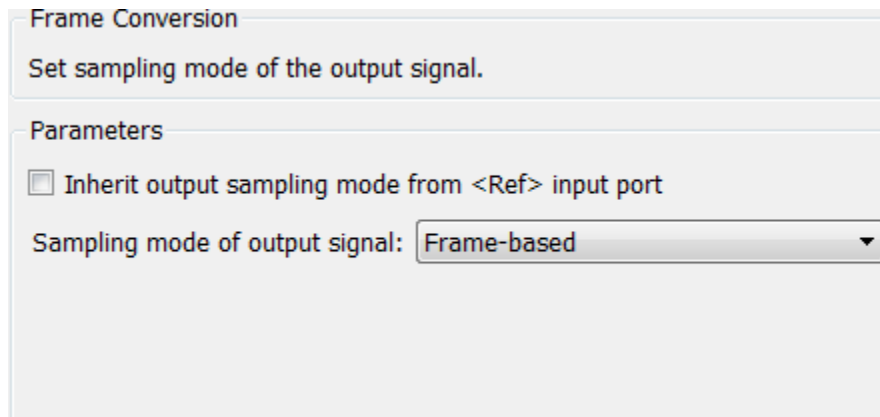


Fig. 6: Simulink block configuration

Then I applied a window function to the input signal for computing the hamming window and the respective block configuration window shown in Fig. 7 below-

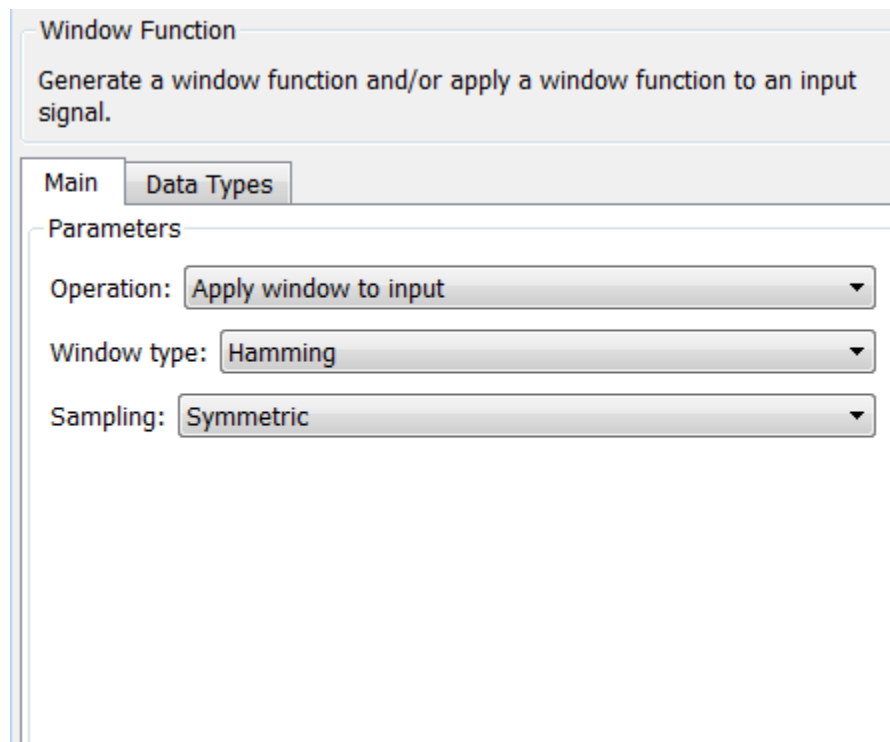


Fig. 7: Simulink block configuration

Thereafter I connected a digital filter block to filter out each channel over time however I configured with IIR type along with defined coefficients and the configuration given in Fig. 8 as-

Digital Filter

Independently filter each channel of the input over time using a specified digital filter implementation. You can specify filter coefficients using either tunable mask dialog parameters or separate input ports, which are useful for time-varying coefficients.

Coefficient source

☒ Dialog parameters

☐ Input port(s)

☐ Discrete-time filter object (DFILT)

Main

Data Types

Parameters

Transfer function type:

IIR (poles & zeros)

Filter structure:

Direct form II transposed

Numerator coefficients:

[1 2]

Denominator coefficients:

[1 0.1]

Input processing:

Columns as channels (frame based)

Initial conditions:

0

View Filter Response

Fig. 8: Simulink block configuration

Further to perform FFT analysis I applied a FFT block and with proper configuration as given in figure 9, I obtained the spectrum representation or frequency domain analysis which I considered in the performance analysis-

FFT

Compute the fast Fourier transform (FFT) across the first dimension of the input.

When you set the 'FFT implementation' parameter to 'Radix-2', the FFT length must be a power of two.

Main **Data Types**

Parameters

FFT implementation: Auto

☐ Output in bit-reversed order

☒ Scale result by FFT length

☒ Inherit FFT length from input dimensions

Fig. 9: Simulink block configuration

Further as per the defined procedure I applied filter bank and computed the log of filter bank energies to get MFCC, however I also applied DCT for expressing a finite number of sequences in terms of summation of cos function at various frequency values. Thus I configured both DCT and IDCT block as depicted in figure 10 & figure 11 below-

DCT

Compute the discrete cosine transform (DCT) across the first dimension of the input. The first dimension of the input must have a power-of-two length.

Main **Data Types**

Fixed-point operational parameters

Rounding mode: Simplest Overflow mode: Wrap

Floating-point inheritance takes precedence over the settings in the 'Data Type' column below. When the block input is floating point, all block data types match the input. When the block input is fixed point, all internal data types are signed fixed point.

	Data Type	Assistant	Minimum	Maximum
Sine table:	Inherit: Same word length as input	>>		
Product output:	Inherit: Inherit via internal rule	>>		
Accumulator:	Inherit: Inherit via internal rule	>>		
Output:	Inherit: Inherit via internal rule	>>	[]	[]

☐ Lock data type settings against changes by the fixed-point tools

Fig. 10: Simulink block configuration

IDCT

Compute the inverse discrete cosine transform (IDCT) across the first dimension of the input. The first dimension of the input must have a power-of-two length.

Main
Data Types

Fixed-point operational parameters

Rounding mode: Floor
Overflow mode: Wrap

Floating-point inheritance takes precedence over the settings in the 'Data Type' column below. When the block input is floating point, all block data types match the input. When the block input is fixed point, all internal data types are signed fixed point.

	Data Type	Assistant	Minimum	Maximum
Sine table:	Inherit: Same word length as input	>>		
Product output:	Inherit: Inherit via internal rule	>>		
Accumulator:	Inherit: Inherit via internal rule	>>		
Output:	Inherit: Inherit via internal rule	>>		

☐ Lock data type settings against changes by the fixed-point tools

Fig. 11: Simulink block configuration

Following the procedure I finally connected a block of audio weighing filter to design an audio weighing filter wherein weighing means the type of frequency response of the filter and for the validation of the design I considered ANSI S1.42 as the standard that I followed in the simulink modeling as mentioned in below figure 12 and respectively filter visualization presented in figure 13-

Audio Weighting Filter
Design an audio weighting filter.

[View Filter Response](#)

Filter specifications

Weighting type: A Class: 1

Impulse response: IIR

Frequency units: Hz Input Fs: 48000

Algorithm

Design method: ANSI S1.42

☒ Scale SOS filter coefficients to reduce chance of overflow

Filter implementation

Structure: Direct-form II SOS

☐ Use basic elements to enable filter customization

☐ Optimize for unit scale values

Input processing: Columns as channels (frame based)

☐ Use symbolic names for coefficients

Fig. 12: Configuration part during Simulink modelling

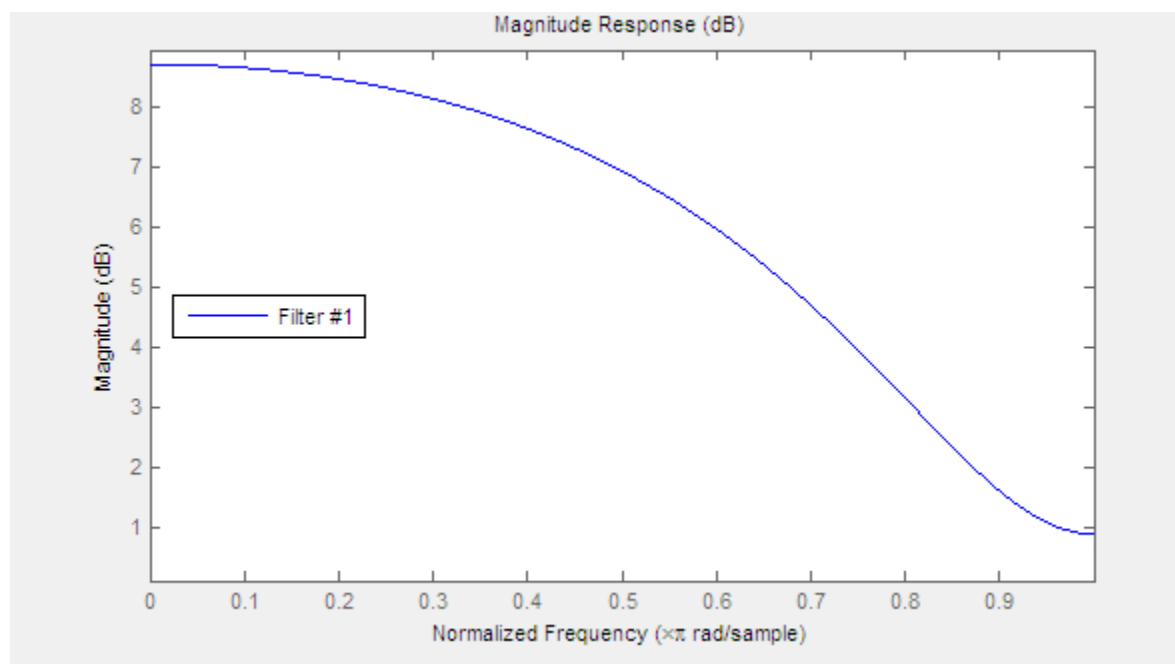
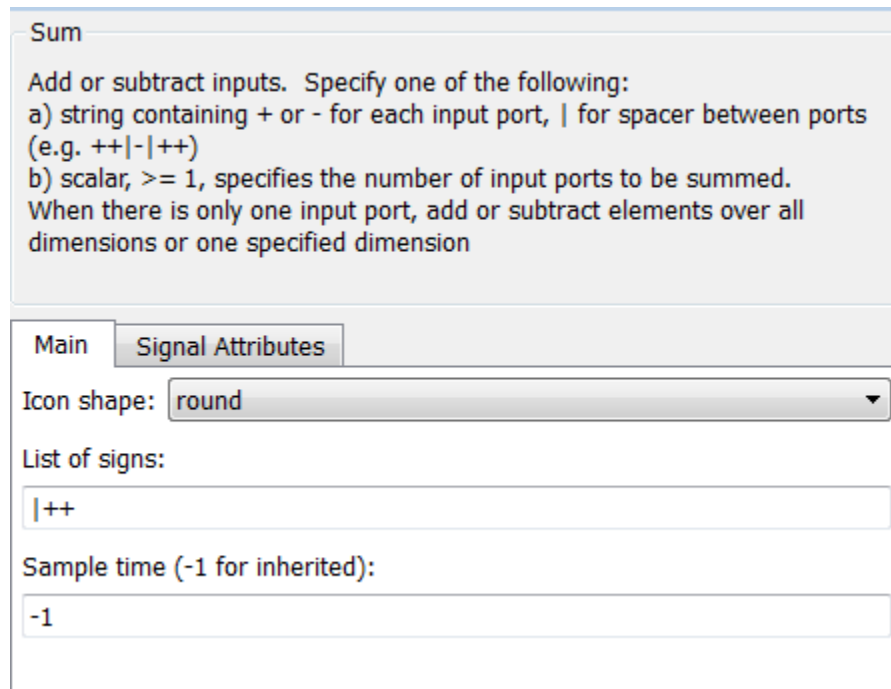


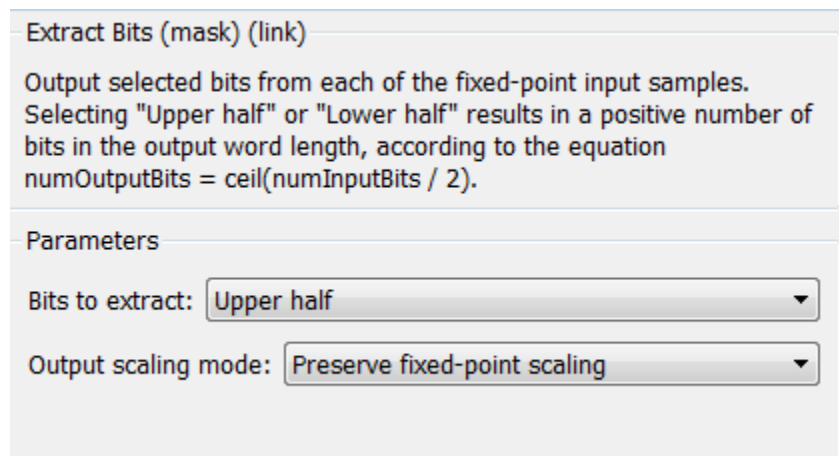
Fig. 13: Simulink block configuration

Similarly other blocks configured which I mentioned in below figures-



The image shows the configuration window for a Simulink 'Sum' block. The title bar says 'Sum'. The main text area contains instructions: 'Add or subtract inputs. Specify one of the following: a) string containing + or - for each input port, | for spacer between ports (e.g. ++|-|++) b) scalar, >= 1, specifies the number of input ports to be summed. When there is only one input port, add or subtract elements over all dimensions or one specified dimension'. Below this, there are two tabs: 'Main' and 'Signal Attributes'. The 'Main' tab is selected. It contains three fields: 'Icon shape:' with a dropdown menu set to 'round', 'List of signs:' with a text input field containing '|++', and 'Sample time (-1 for inherited):' with a text input field containing '-1'.

Fig. 14: Simulink block configuration



The image shows the configuration window for a Simulink 'Extract Bits (mask) (link)' block. The title bar says 'Extract Bits (mask) (link)'. The main text area contains instructions: 'Output selected bits from each of the fixed-point input samples. Selecting "Upper half" or "Lower half" results in a positive number of bits in the output word length, according to the equation $\text{numOutputBits} = \text{ceil}(\text{numInputBits} / 2)$.' Below this, there are two tabs: 'Parameters' and 'Signal Attributes'. The 'Parameters' tab is selected. It contains two fields: 'Bits to extract:' with a dropdown menu set to 'Upper half', and 'Output scaling mode:' with a dropdown menu set to 'Preserve fixed-point scaling'.

Fig. 15: Simulink block configuration

In this manner I completed the simulink modeling part in the project and by following proper procedure I discussed the performance analysis wherein different plots presented.

CE 2.7

For my research study I investigated these steps in detail and in performance analysis I referred to the simulink model that was created as per the procedure in MATLAB, however due to constant variation in the audio signal, things need to be simplified so I made use of some assumptions like one could assume not many variations in the audio signals for a short time scale. In the case of the shorter frame, there would be not enough samples for getting the desired spectral estimate and in the case of the longer frame, the signal varies a lot throughout the frame. In the next step I carried out the computation of the frame power spectrum whereas it mainly depends on the human cochlea which vibrates at distinct spots based on incoming sound frequency. The distinct nerves inform the brain about the presence of certain frequencies based on the vibration of location in the cochlea. Further I studied that the periodogram performs a similar operation, identification of the presence of frequencies in the frame. The spectral estimate of the periodogram comprises still irrelevant information for the ASR. Therefore, some periodogram bin clumps could be taken and summed up for getting clarity about the existence of energy in distinct frequency regions. For this, I studied that Mel filterbank is utilized and technically I understood that this filter indicates the energy which exists. Further, I studied about the Mel scale which helps determine how to provide spacing to filter banks and how to make them wider. For finding the filterbank energies, a logarithm of the same needs to be taken. I investigated that the logarithm enables utilization of the channel normalization technique, Cepstral mean subtraction. The next step involves the computation of log filterbank energies DCT. I studied the reasons behind its performance. This step is mainly done because of overlapping filterbanks. For energies de-correlation, DCT is used, and diagonal covariance matrices could be utilized for modeling features. By dropping the DCT coefficients, I studied that a small improvement could be obtained in the performance of ASR. Here, I completed a study of associated steps and obtained useful information. For analyzing the performance of the developed model I worked in the MATLAB environment wherein I also referred to the standard blocks and functions to process the speech signal as per the standard procedure. Therefore in the below figures I illustrated the MFCC application in the speech recognition and various plots obtained for analyzing the system performance-

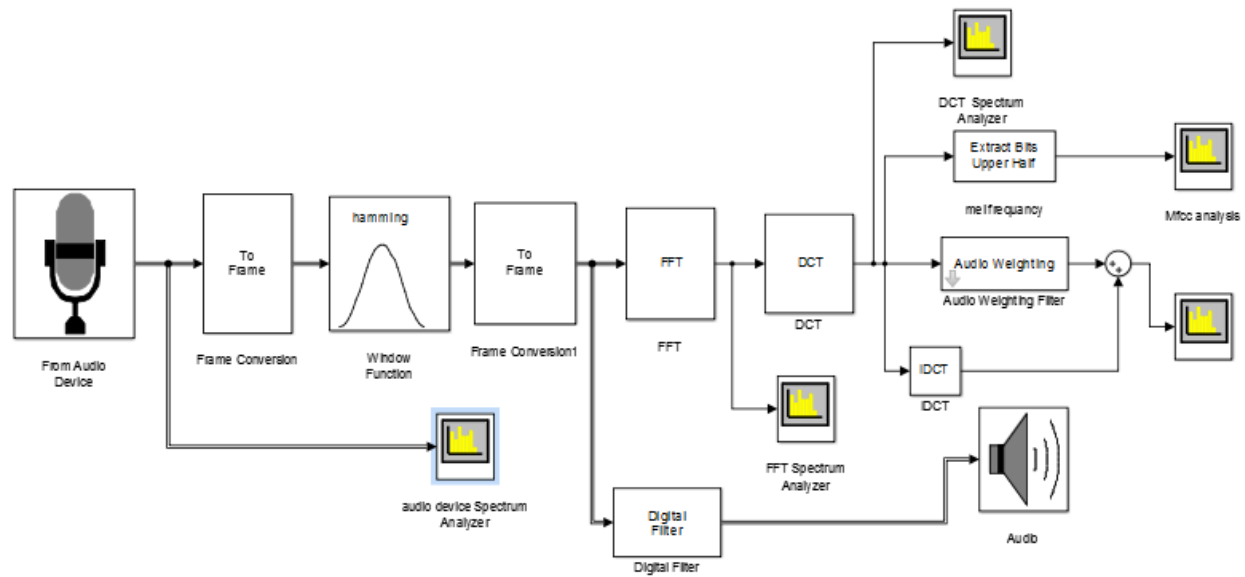


Fig. 16: Simulink model

In order to analyze the signal I utilized the spectrum analyzer at different blocks i.e. shown in above figure 16 such as Audio device spectrum analyzer used for input audio device spectrum. More significantly I followed standard configuration procedure for the FFT spectrum analyzer to conduct FFT signal analysis as depicted in figure 17 below-

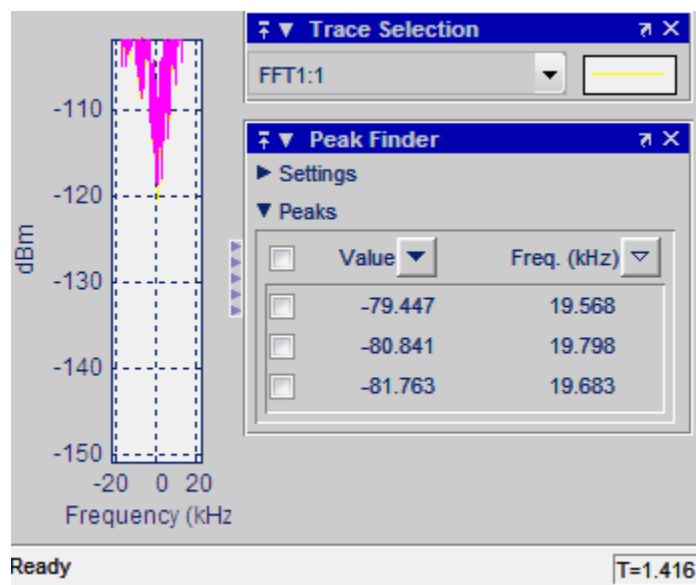


Fig. 17: Simulink block configuration

In the similar manner I configured the MFCC spectrum analyzer for MFCC analysis as shown in figure 18 below-

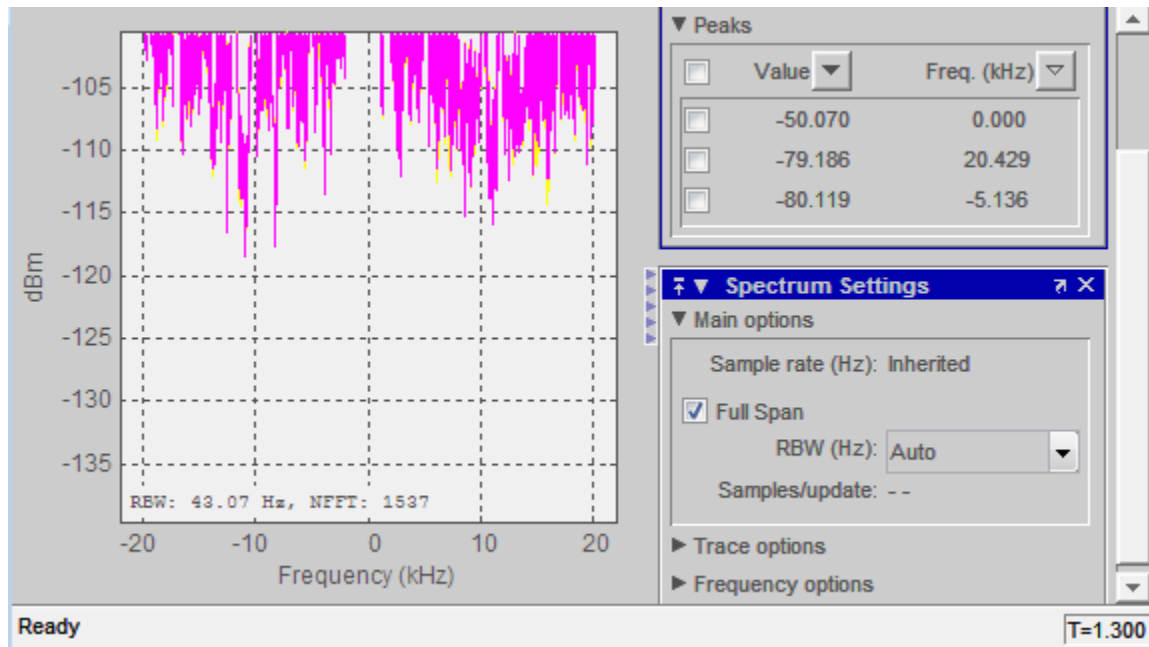


Fig. 18: Simulink block configuration

Finally corresponding to the different spectrum analyzer I obtained the plots that I shown as follows-

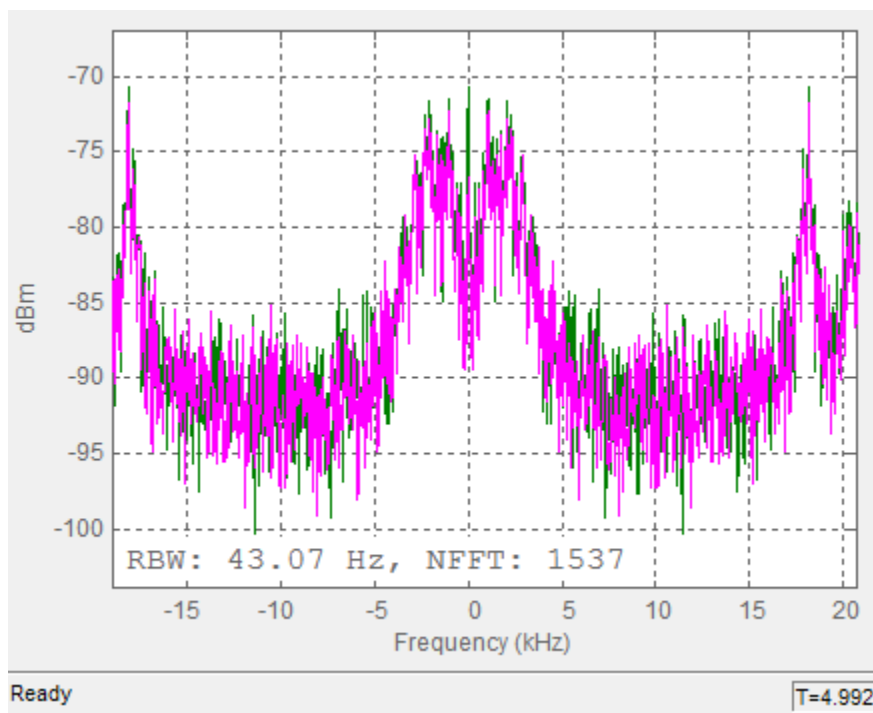


Fig. 19: Input Audio device spectrum

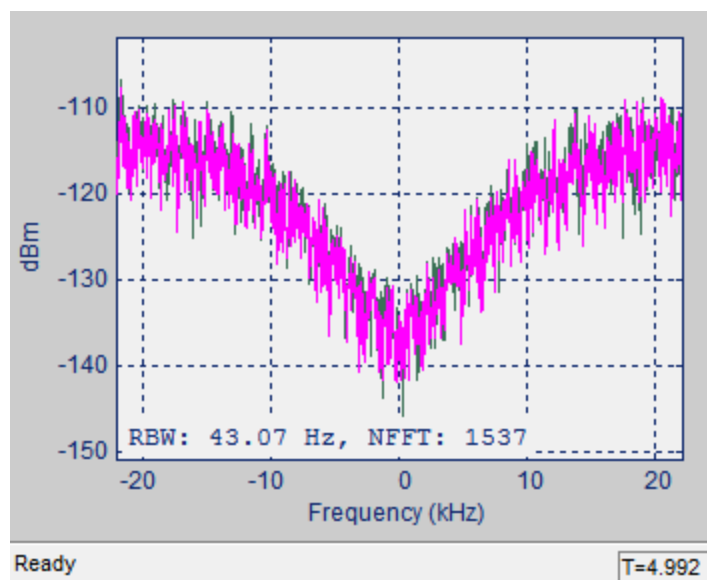


Fig. 20: FFT analysis

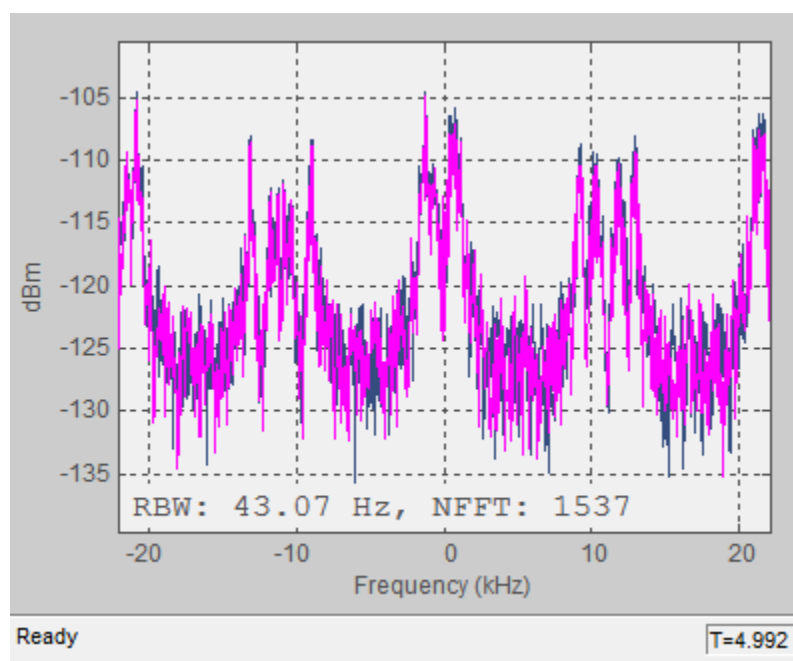


Fig. 21: Mel filter spectrum

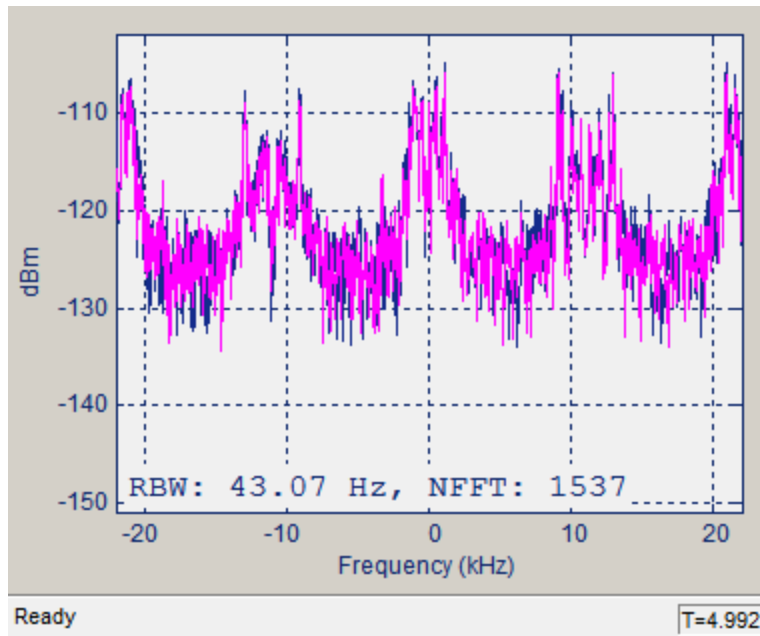


Fig. 22:DCT analysis

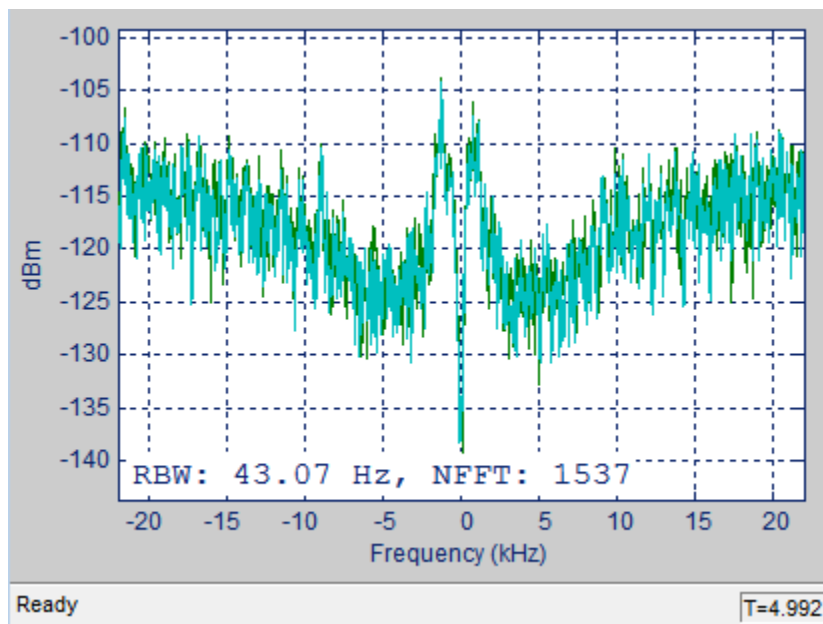


Fig. 23:Audio Weighting filter output

Overall through the presented modeling analysis of a speech recognition using MFCC application I executed all the significant steps involved in the process such as audio input signal, short duration energy analysis, window function applied, extraction of characteristics constraints using FFT, filter bank analysis, log computation of filter bank energies followed by DCT and

audio weighting filter. More significantly I implemented multiple spectrum analyzers at different steps to analyze the performance of the model.

CE 2.8

The challenging task for me in this project was the research review on existing speech recognition algorithms. To overcome this challenge I first discussed with my supervisor regarding selection of literature contents under the research review analysis and my supervisor suggested indexed journals only, so I found relevant research journals and reviewed the literature contents. Based on research review I intentionally studied and recorded the applications of MFCCs along with detailed information concerning steps efficiently without complexities. I systematically recorded steps and ensured it to be in proper sequence. I followed all the steps in the modeling analysis which I performed in MATLAB. I optimally studied the application of MFCCs in the speech recognition system and acquired the desired information without problems. For gathering information concerning the project with accuracy and overcoming research-related issues, I used research aptitude proficiently and referred to relevant literature contents.

D. SUMMARY

CE 2.9

I made diligent efforts to finish the given project with efficacy and as per the stipulated time period. Initially, I gained knowledge of project-related concepts and scope in detail. I put my technical knowledge to use while applying the technologies and perfectly executed the project tasks. To choose and deploy the best methodology while performing the activities, a thorough investigation was carried out by me. I assured the accomplishment of work by optimally following the prepared action plans. I adopted corrective measures considering the risk factors underlying during the performance of activities. The information collected by studying and analyzing the several textbooks was recorded in a report along with the information of research conducted about the simplified strategies. I employed my critical thinking ability as well as practical knowledge for identifying and resolving the obstacles that occurred while conducting the activities. The project was a crucial opportunity for me to improvise my skills and to utilize my theoretical knowledge practically. The duties assigned by my project supervisor were

fulfilled very responsibly and I excellently submitted the work. I punctually worked and successfully accomplished the project within the given timeframe which was applauded by my departmental head. My contextual knowledge was enhanced by completing the project and I also gained creative skills.

CE 2.10

Through present work, I studied MFCCs application in speaker or speech recognition, however speaker recognition could simplify translating speech tasks in systems which are trained on the voice of the concerned person. For analyzing the research study I followed all the steps in the modeling analysis which I performed in MATLAB. This could be utilized for authenticating or verifying the speaker's identity as a security process part. I studied that the speaker recognition system includes two phases namely verification and enrollment. In the automated speech recognition system, the initial step is the extraction of features through which the audio signal components could be identified which forms the basis for the identification of linguistic content and elimination of other stuff. I properly studied the application of the MFCCs in systems associating speech recognition. The MFCCs help in recognizing speakers or people from their voices. I studied that MFCCs are widely and increasingly utilized in applications concerning music information retrieval like the audio similarity measures and others. I completed research work proficiently and submitted a research project.